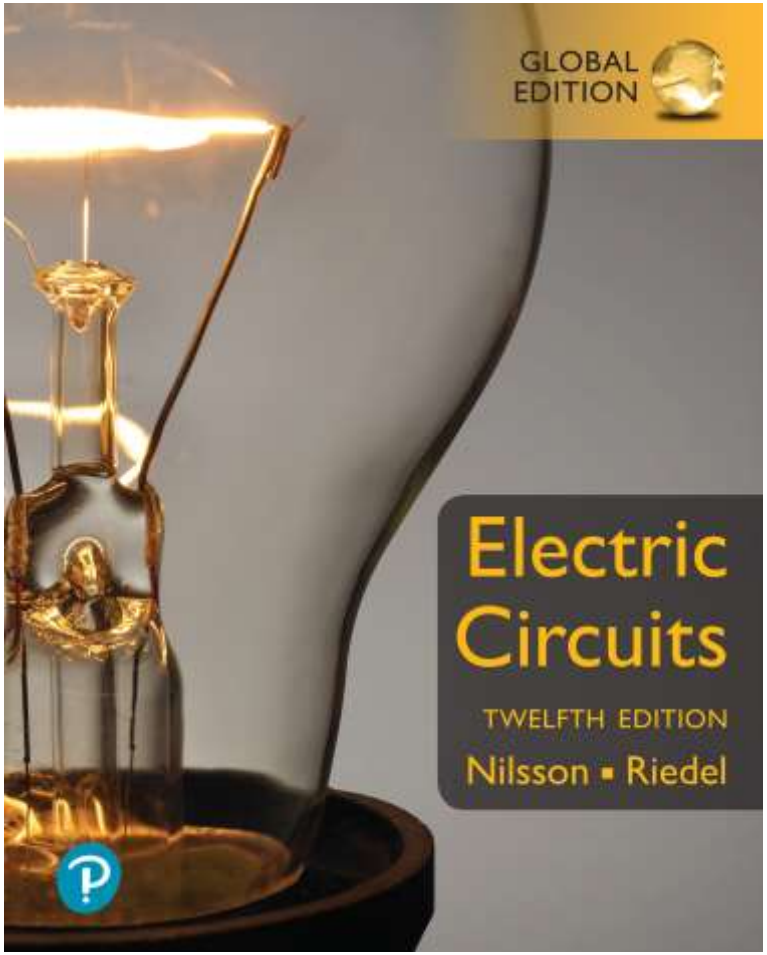
Electric Circuits

Twelfth Edition, Global Edition



Chapter 5 – Part 1

The Operational Amplifier

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Objectives

5.1 Be able to identify and name the terminals of an op amp

5.2 Be able to state the constraints that characterize an ideal op amp

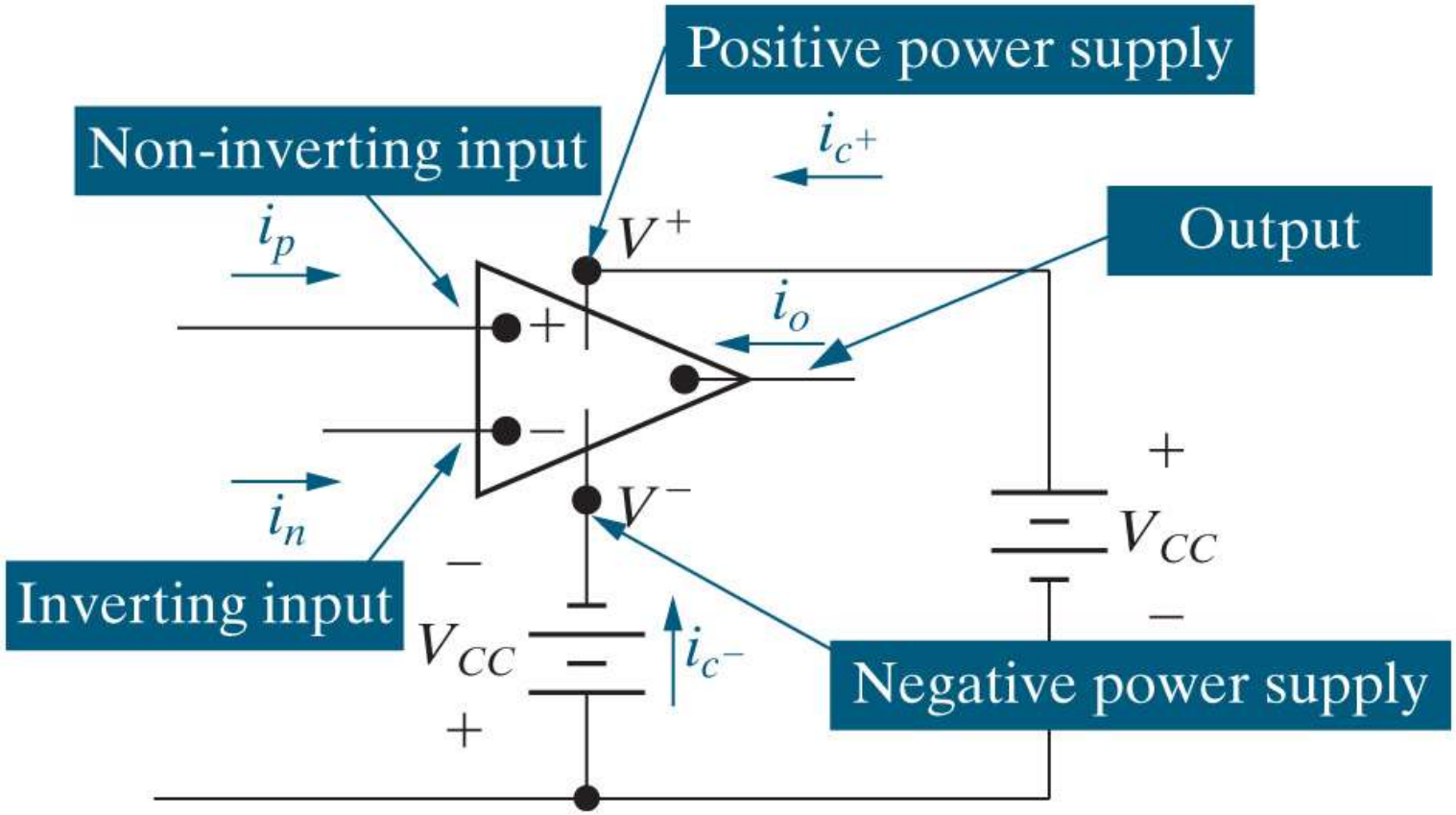
5.3 Know how to recognize and analyze the following op amp circuits: inverting amplifier, summing amplifier, non-inverting amplifier, difference amplifier

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Chapter 5 – Op Amps (1 of 22)

Terminal names, voltages and currents

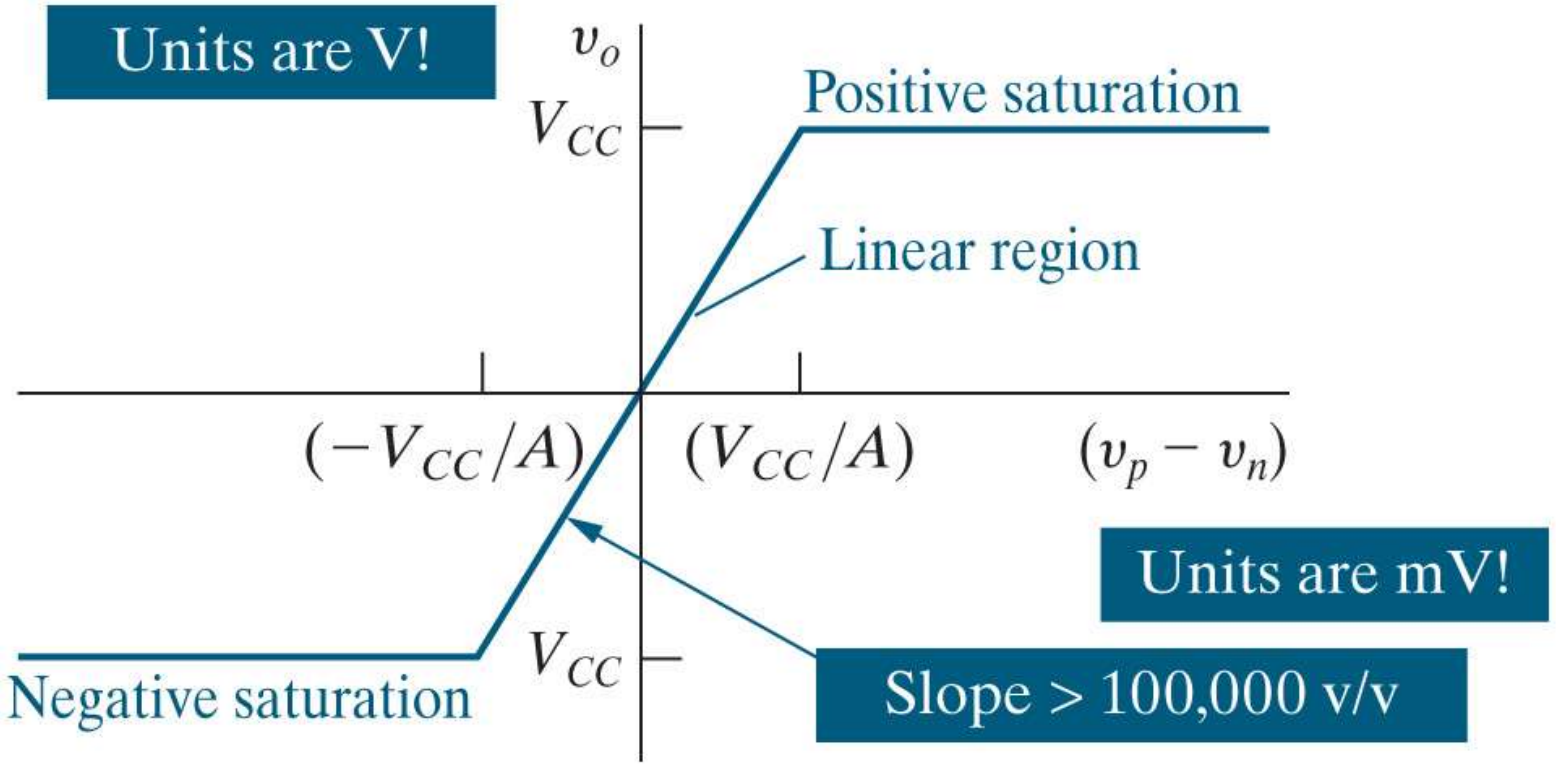


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Chapter 5 – Op Amps (2 of 22)

Voltage transfer characteristic



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Chapter 5 – Op Amps (3 of 22)

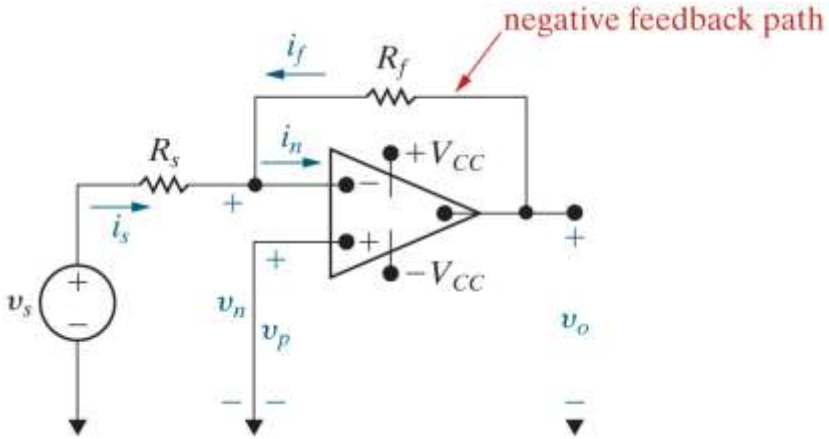
Op amps are inherently linear devices.

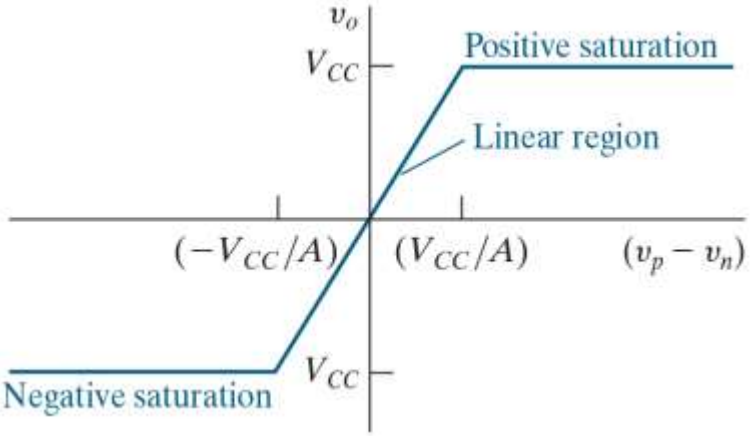
- ✗ A. True
- ✓ B. False

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Chapter 5 – Op Amps (4 of 22)

- Since this class only deals with linear circuits, we use op amps only when they are operating in their linear region.
- How can we configure an op amp to operate in its linear region? Negative feedback!





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Chapter 5 – Op Amps (5 of 22)

- How can we be sure that the op amp is operating in its linear region? We can't! We must assume that it is operating in its linear region, complete our circuit analysis, and check our assumption. If the op amp is operating in its linear region,

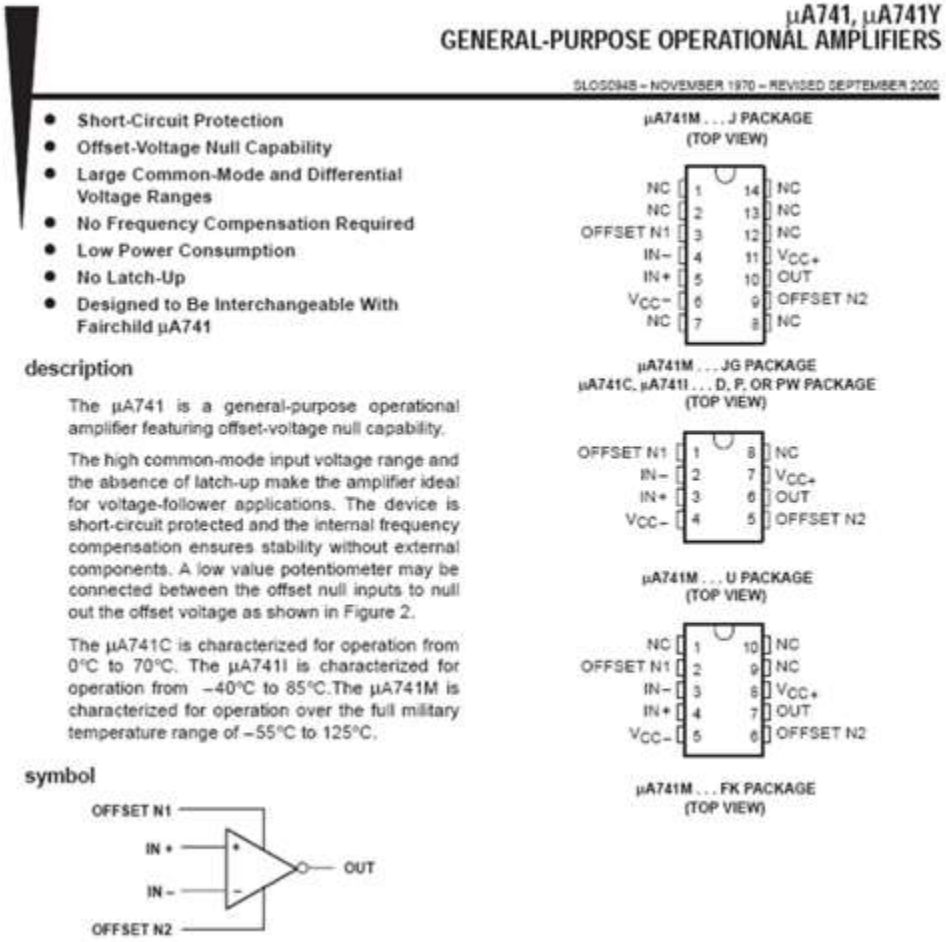
$$-V_{CC} < v_o < V_{CC}$$

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Chapter 5 – Op Amps (6 of 22)

Texas Instruments μA741 op amp data sheet



μA741, μA741Y
GENERAL-PURPOSE OPERATIONAL AMPLIFIERS

SLOS094B – NOVEMBER 1970 – REVISED SEPTEMBER 2000

- Short-Circuit Protection
- Offset-Voltage Null Capability
- Large Common-Mode and Differential Voltage Ranges
- No Frequency Compensation Required
- Low Power Consumption
- No Latch-Up
- Designed to Be Interchangeable With Fairchild μA741

description

The μA741 is a general-purpose operational amplifier featuring offset-voltage null capability.

The high common-mode input voltage range and the absence of latch-up make the amplifier ideal for voltage-follower applications. The device is short-circuit protected and the internal frequency compensation ensures stability without external components. A low value potentiometer may be connected between the offset null inputs to null out the offset voltage as shown in Figure 2.

The μA741C is characterized for operation from 0°C to 70°C. The μA741I is characterized for operation from −40°C to 85°C. The μA741M is characterized for operation over the full military temperature range of −55°C to 125°C.

symbol

μA741M . . . J PACKAGE
(TOP VIEW)

μA741M . . . JG PACKAGE
μA741C, μA741I . . . D, P, OR PW PACKAGE
(TOP VIEW)

μA741M . . . U PACKAGE
(TOP VIEW)

μA741M . . . FK PACKAGE
(TOP VIEW)

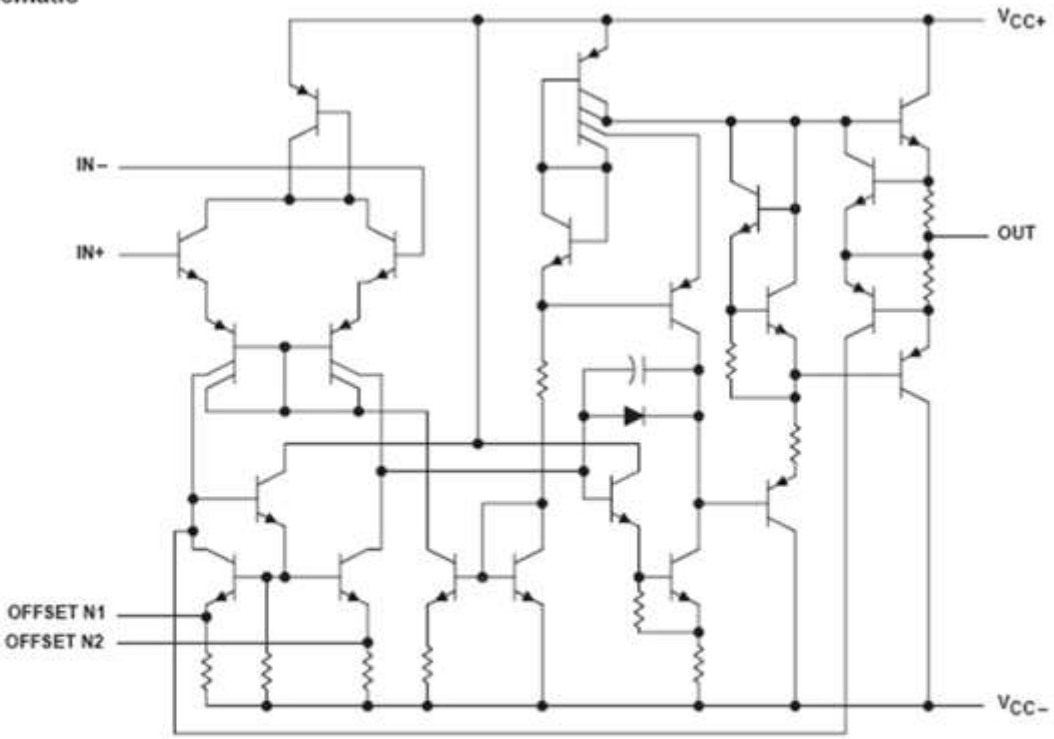
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Chapter 5 – Op Amps (7 of 22)

TI μA741 op amp data sheet (continued)

schematic



Component Count	
Transistors	22
Resistors	11
Diode	1
Capacitor	1

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Chapter 5 – Op Amps (8 of 22)

TI μA741 op amp data sheet (continued)

electrical characteristics at specified free-air temperature, $V_{CC\pm}$ = ±15 V (unless otherwise noted)

PARAMETER		TEST CONDITIONS	T_A†	μA741C			μA741I, μA741M			UNIT
				MIN	TYP	MAX	MIN	TYP	MAX	
V_{IO}	Input offset voltage	V_O = 0	25°C		1	6		1	5	mV
			Full range			7.5			6	
$\Delta V_{IO(adj)}$	Offset voltage adjust range	V_O = 0	25°C		±15			±15		mV
I_{IO}	Input offset current	V_O = 0	25°C		20	200		20	200	nA
			Full range			300			500	
I_{IB}	Input bias current	V_O = 0	25°C		80	500		80	500	nA
			Full range			800			1500	
V_{ICR}	Common-mode input voltage range		25°C	±12	±13		±12	±13		V
			Full range	±12			±12			
V_{OM}	Maximum peak output voltage swing	R_L = 10 kΩ	25°C	±12	±14		±12	±14		V
		R_L ≥ 10 kΩ	Full range	±12			±12			
		R_L = 2 kΩ	25°C	±10	±13		±10	±13		
		R_L ≥ 2 kΩ	Full range	±10			±10			
A_{VD}	Large-signal differential voltage amplification	R_L ≥ 2 kΩ	25°C	20	200		50	200		V/mV
		V_O = ±10 V	Full range	15			25			
r_i	Input resistance		25°C	0.3	2		0.3	2		MΩ
r_o	Output resistance	V_O = 0, See Note 5	25°C		75			75		Ω
C_i	Input capacitance		25°C		1.4			1.4		pF
CMRR	Common-mode rejection ratio	V_{IC} = V_{ICR}min	25°C	70	90		70	90		dB
			Full range	70			70			
k_{SVS}	Supply voltage sensitivity ($\Delta V_{IO}/\Delta V_{CC}$)	V_{CC} = ±9 V to ±15 V	25°C		30	150		30	150	μV/V
			Full range			150			150	
I_{OS}	Short-circuit output current		25°C		±25	±40		±25	±40	mA
I_{CC}	Supply current	V_O = 0, No load	25°C		1.7	2.8		1.7	2.8	mA
			Full range			3.3			3.3	
P_D	Total power dissipation	V_O = 0, No load	25°C		50	85		50	85	mW
			Full range			100			100	

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Chapter 5 – Op Amps (9 of 22)

In a uA741 op amp, the voltage gain in the linear operating region is 200,000 V/V. In an ideal op amp, the voltage gain in the linear region is

- ✗ A. About the same
- ✗ B. Much smaller
- ✗ C. 0
- ✓ D. ∞

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Chapter 5 – Op Amps (10 of 22)

In a uA741 op amp, the input resistance is about $2\text{M}\,\Omega$.

In an ideal op amp, the input resistance is

- ✗ A. About the same
- ✗ B. Much smaller
- ✗ C. 0
- ✓ D. ∞

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Chapter 5 – Op Amps (11 of 22)

In a uA741 op amp, the output resistance is about 75 Ω.

In an ideal op amp, the output resistance is

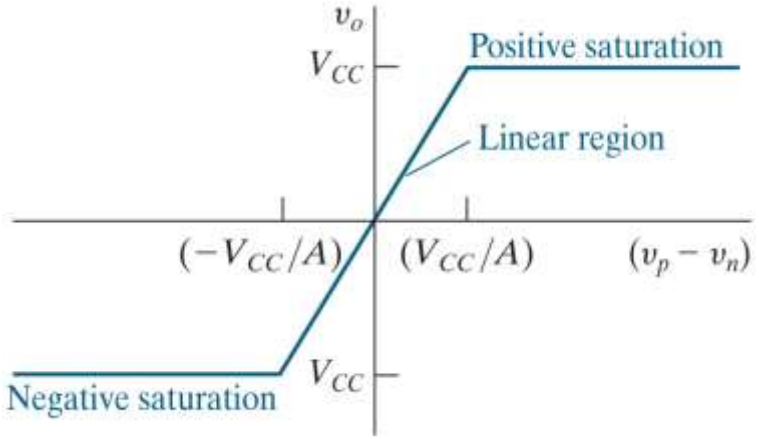
- ✗ A. About the same
- ✗ B. Much larger
- ✓ C. 0
- ✗ D. ∞

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Chapter 5 – Op Amps (12 of 22)

In an ideal op amp, the voltage gain in the linear region is infinite. Therefore, what is $(v_p - v_n)$?



- ✔ A. 0
- ✘ B. 200,000
- ✘ C. V_{CC}
- ✘ D. ∞

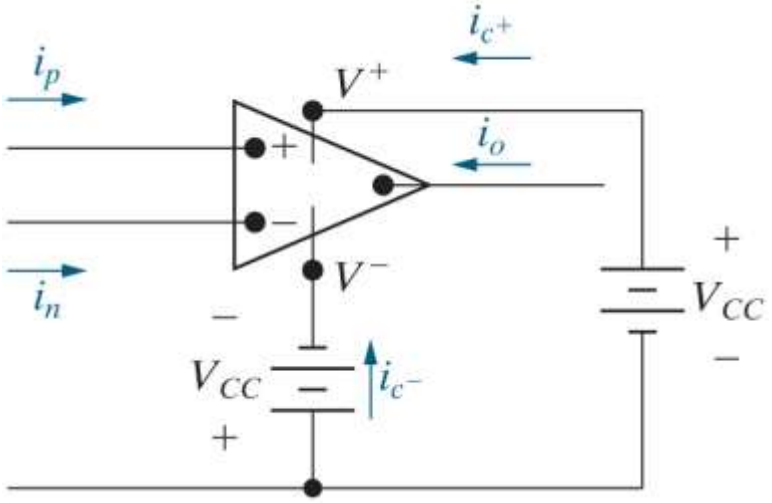
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Chapter 5 – Op Amps (13 of 22)

In an ideal op amp, the input resistance is infinite. Therefore, the currents flowing into the input of the op amp $(i_p \text{ and } i_n)$ are

- ✗ A. Both infinite
- ✓ B. Both zero
- ✗ C. Determined by circuit analysis
- ✗ D. Undefined



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Chapter 5 – Op Amps (14 of 22)

We usually assume the op amp is ideal and operating in its linear region. This allows us to make two important assumptions that simplify the circuit analysis:

- The voltages at the inverting and non-inverting input terminals are the same with respect to a common reference node. That is, $v_p = v_n$.
- The currents entering the inverting and non-inverting input terminals are both zero. That is, $i_p = i_n = 0$.

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Chapter 5 – Op Amps (15 of 22)

The analysis method for circuits with ideal op amps:

- Step 1 – Check for a negative feedback path. If it exists, assume the op amp operates in its linear region.
- Step 2 – Write a KCL equation at the inverting input terminal.
- Step 3 – Solve the KCL equation and use the solution to find the op amp's output voltage.
- Step 4 – Compare the op amp's output voltage to the power supply voltages to determine if the op amp is operating in its linear region or if it is saturated.

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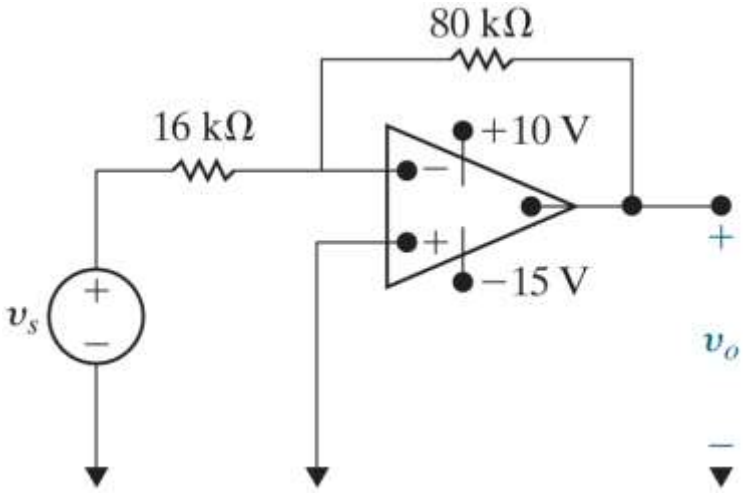
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Chapter 5 – Op Amps (16 of 22)

Example

Find v_o for $v_S = 0.4\,\text{V}, 2.0\,\text{V}, 3.5\,\text{V}, -0.6\,\text{V}, -1.6\,\text{V},$ and $-2.4\,\text{V}.$

Assume the op amp is ideal.



- Step 1 – Check for a negative feedback path. It exists, so assume the op amp operates in its linear region.
- Step 2 – Write a KCL equation at the inverting input terminal. Use the op amp assumptions that $v_p = v_n$ and $i_p = i_n = 0.$

$$v_p = 0 \Rightarrow v_n = v_p = 0$$

$$\text{KCL at } v_n: \frac{0 - v_s}{16{,}000} + \frac{0 - v_o}{80{,}000} + 0 = 0$$

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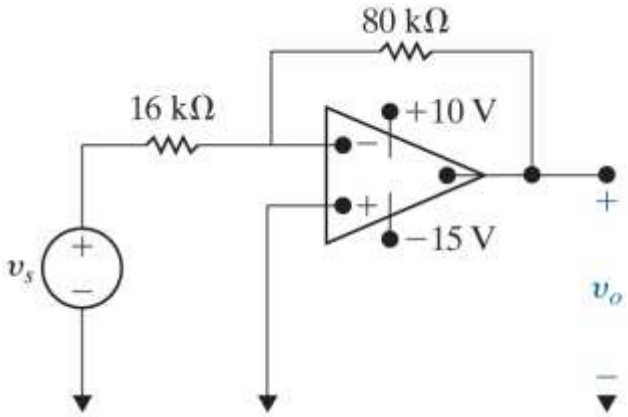
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Chapter 5 – Op Amps (17 of 22)

Example

Find v_o for $v_S = 0.4\,\text{V}, 2.0\,\text{V}, 3.5\,\text{V}, -0.6\,\text{V}, -1.6\,\text{V},$ and $-2.4\,\text{V}.$

Assume the op amp is ideal.



- Step 3 – Solve the KCL equation and use the solution to find the op amp's output voltage.

$$v_o = -\frac{80{,}000}{16{,}000} v_s = -5v_s$$

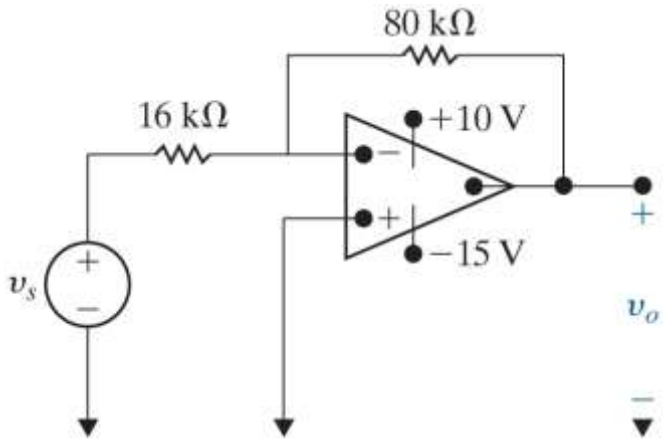
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Chapter 5 – Op Amps (18 of 22)

Example

Find v_o for $v_S = 0.4\,\text{V}, 2.0\,\text{V}, 3.5\,\text{V}, -0.6\,\text{V}, -1.6\,\text{V},$ and $-2.4\,\text{V}$. Assume the op amp is ideal.



- Step 4 – Compare the op amp's output voltage with the power supply voltages to determine if the op amp is operating in its linear region or if it is saturated. The results are shown in the table.

v_s [V]	$v_o = -5v_s$ [V]	OK?
0.4	–2.0	Yes
2.0	–10.0	Yes
3.5	–17.5	NO – saturate at −15 V
–0.6	3.0	Yes
–1.6	8.0	Yes
−2.4	12.0	NO – saturate at 10 V

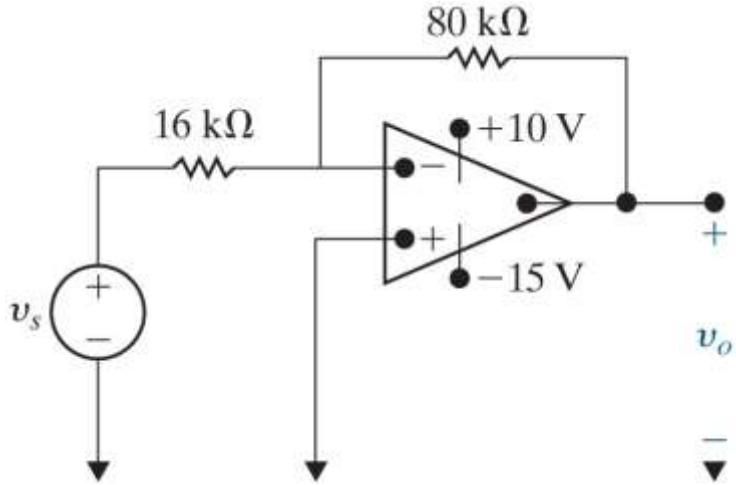
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Chapter 5 – Op Amps (19 of 22)

Example, continued

Find the range of input voltages v_s for which the op amp operates in its linear region.



$$v_o = -5v_s = 10 \quad \Rightarrow \quad v_s = -2 \text{ V}$$

$$v_o = -5v_s = -15 \quad \Rightarrow \quad v_s = 3 \text{ V}$$

$$\therefore \quad -2 \text{ V} \le v_s \le 3 \text{ V}$$

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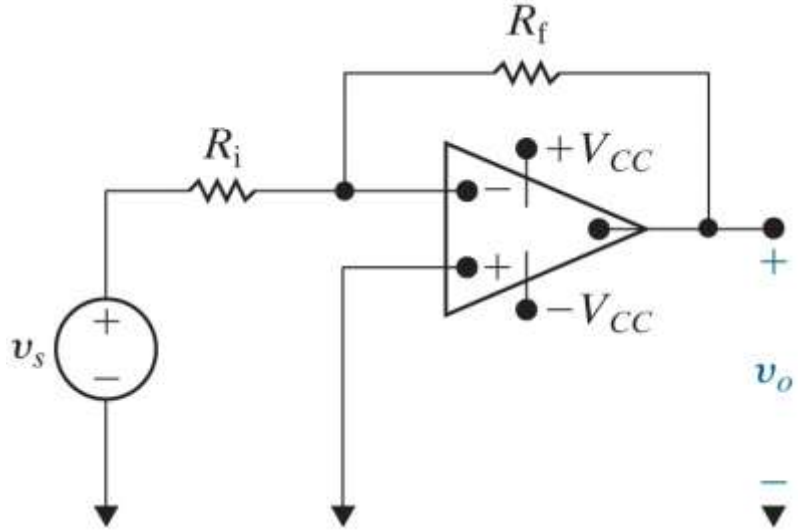
Chapter 5 – Op Amps (20 of 22)

Inverting Amplifier

$$v_p = 0 \Rightarrow v_n = v_p = 0$$

$$\text{KCL at } v_n: \frac{0 - v_s}{R_i} + \frac{0 - v_o}{R_f} + 0 = 0$$

$$\therefore \qquad v_o = -\frac{R_f}{R_i} v_s$$



Design equation:

$$v_o = -\frac{R_f}{R_i} v_s \qquad \text{only if} \qquad -V_{CC} \le v_o \le +V_{CC}$$

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Chapter 5 – Op Amps (21 of 22)

In describing op amp circuits, the term “inverting” means

- ✓ A. Changing the sign
- ✗ B. Operating linearly
- ✗ C. 1/(the input)
- ✗ D. All of the above

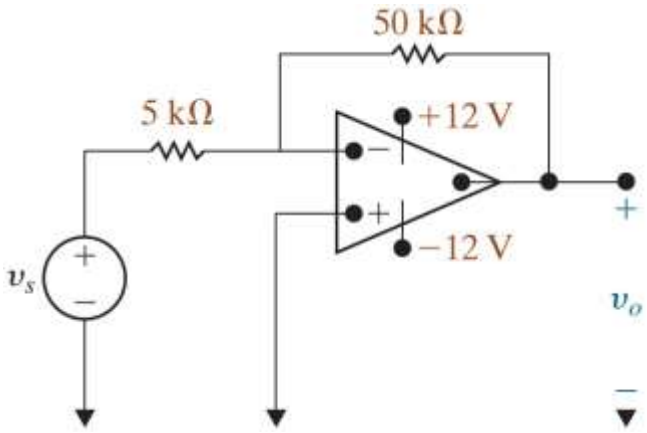
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Chapter 5 – Op Amps (22 of 22)

Design an inverting amplifier with a gain of 10. Use $\pm 12\,\text{V}$. power

Choose a feedback resistor and input resistor such that their ratio is 10 – one possibility is shown below:



$$v_o = -10v_s = 12 \quad \Rightarrow \quad v_s = -1.2\ \text{V}$$

$$v_o = -10v_s = -12 \quad \Rightarrow \quad v_s = 1.2\ \text{V}$$

$$\therefore \quad v_o = -10v_s \quad \text{for} \quad -1.2\ \text{V} \le v_s \le 1.2\ \text{V}$$

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